

Abstract:

The invention and popularization of Large Language Models(LLMs) has greatly lowered the threshold of difficulty for a large variety of activities involving coding, writing, and online interaction. This decrease in difficulty may lead to an increase in those activities, a substantial portion of which will be LLM generated. This may cause a significant problem in training future models as it will be difficult to separate content generated by humans from content generated by LLMs, as well as decreasing author discoverability and possibly resulting in a homogenisation of style. In this paper, we analyze the activity on online writing website Scribblehub from its inception at the end of 2018 till the end of 2025 and look for anomalies that may be the result of authors using AI assistance. Since it is not possible to directly ascertain whether or not a fiction was written with the assistance of an LLM, this paper focuses on general, sitewide trends. The initial popularization of LLMs like ChatGPT did not seem to have any significant impact on Scribblehub, but in early 2025 there was a 42% increase in the average number of fictions published per month compared to what would have been expected from previous months. This increase presumably continues to the present day, though data from 2026 was not collected. This increase is concentrated among fictions with either 0 ratings, or an average rating of 5 stars, suggesting that the increase in fictions is not concurrent with an increase in readers, or that there is a discoverability problem on the site. The increase is also concentrated among fictions of the Psychological, Mystery, Tragedy, and Drama genres, while the Girls Love, Ecchi, Fanfiction, Harem, and Smut(Typically associated with erotica) genres were barely affected.

Introduction

The invention and development of Large Language Models (LLMs) has greatly decreased the difficulty of producing large volumes of readable, coherent text. This has substantially changed many aspects of the internet and how people interact with it. Many studies have been done on increases in the number of bots that attempt to control discourse, and a decrease in the number of posts on knowledge sharing websites such as Stack Overflow. This paper aims to analyze the adoption of LLMs among creatives, absent other incentives such as money or controlling discourse. Here we analyze the effect that this developing technology has had on online writing communities. Specifically, this paper focuses on Scribblehub, a website where authors can publish their stories for anyone to read and give feedback. While previous research has been focused on goal oriented internet tasks, this website is more recreational, with minimal profit potential. There is no direct monetary reward for authors from the site, although many have a Patreon where fans can directly donate. These stories are also not an information source for their readers, and there is little for powerful groups like governments or corporations to gain by influencing them. These factors make Scribblehub a good source for determining the use of LLMs by the general public for creative tasks. This paper analyzes site-wide trends in the number of fictions published and stylometric differences and finds evidence that LLM adoption among authors on Scribblehub substantially increased in March 2025, which coincides with several advances in LLM's ability to write long form text. Future research might check if these findings are generalizable to all online writing communities, and provide more evidence confirming these findings.

Methods

Data Collection

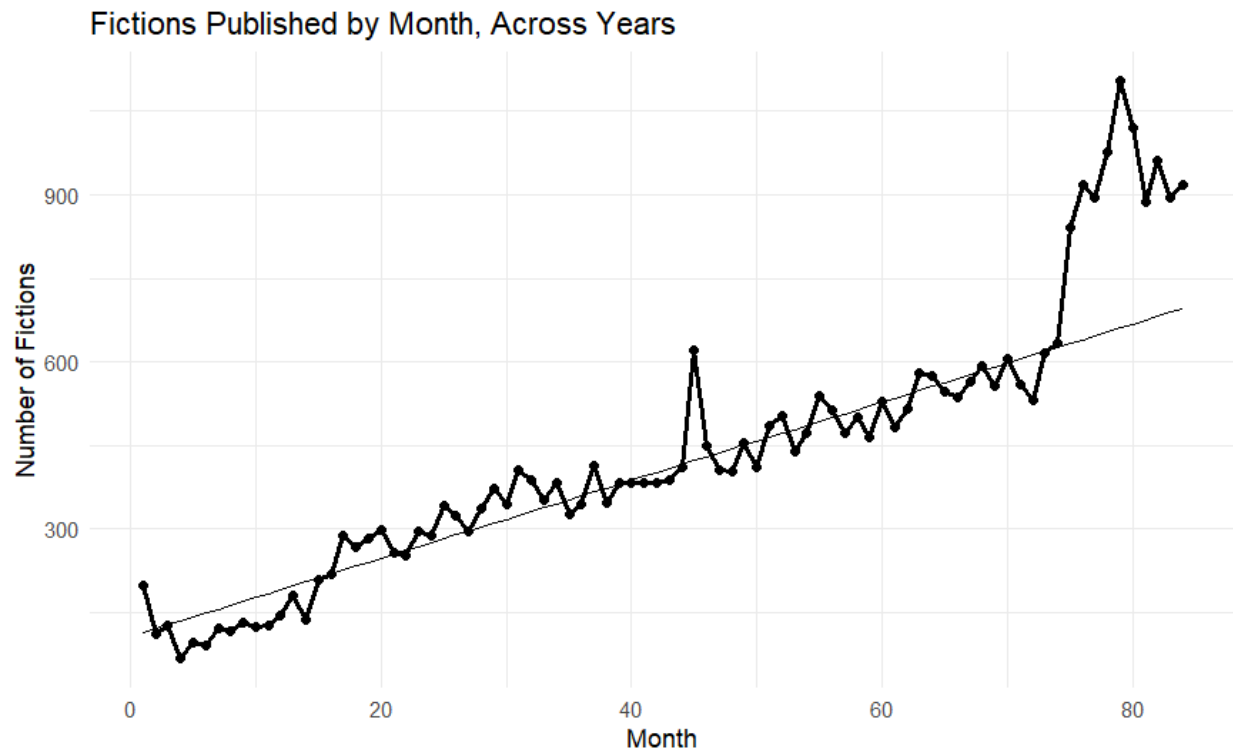
Because it is not possible to reliably determine what literature is written by or with the assistance of LLMs, this paper uses data collected from all of the fictions published to Scribblehub from its inception in late 2018 until the end of 2025 and analyzes trends that may be related to an increase in the use of artificial intelligence by authors. We looked at changing trends in publication volume, Genre concurrence networks, and

relative word frequency that could be correlated with increased LLM use. This method is useful given the inability to directly tell what is LLM assisted and what is not, but means that the data and conclusions of this paper are not necessarily caused by increased LLM use. Due to time constraints, only data from Scribblehub was able to be collected for analysis.

We scraped directly from Scribblehub, collecting data on approximately 37,000 fictions. For each fiction, we extracted the Title, Author, Synopsis, Genre, Tags, Warnings, Reviews, Text from the first chapter, Time published, Average Rating, Number of people who rated, and the link to that fiction's page. A small number of fictions were not collected due to errors in the scraping process, but the distribution of them is approximately random, and the number is less than 1% of the total fictions on the site that were published before 2025. Scraping was done using the Chromote package in R to simulate a Chrome browser. There was a waiting period of 10 seconds in between page requests in order to not burden the site. Scribblehub does not have any provisions against scraping in its TOS.

After being collected, the data was analyzed for anomalies in the number of fictions, ratings of fictions, and genres of fictions, across the publishing time. We first visually inspected monthly publication counts over time. This revealed an apparent structural break beginning in early 2025. We therefore fit linear regression separately to the pre-break and post-break periods to characterize this shift. The data was also analyzed by using the relative word frequencies of the first chapter with the “Stylo” package in R.

Fig 1



Results and Discussion

The public release of LLMs in the winter of 2022 did not have a substantial impact on the fictions published to Scribblehub based on the data. In fact, there was no observable change in the volume of fictions published until March 2025, the number of fictions published increased by approximately 42% and stayed consistently high for at least the rest of the year. The timing of this spike somewhat conflicts with the LLM assistance hypothesis, but there are multiple possible explanations for this. Perhaps there is a generational element to the adoption of AI assistance, where older authors were resistant to using LLM assistance, but a newer generation doesn't care as much. As these tools increase in quality and power every year, perhaps authors who previously have considered them not good enough have changed their minds.

The impact of this sudden change was not uniform across different genres. The Psychological, Mystery, Tragedy, and Drama genres had particularly high increases of 105%, 95%, 88%, and 82% respectively. The Girls Love, Ecchi, Fanfiction, Harem, and

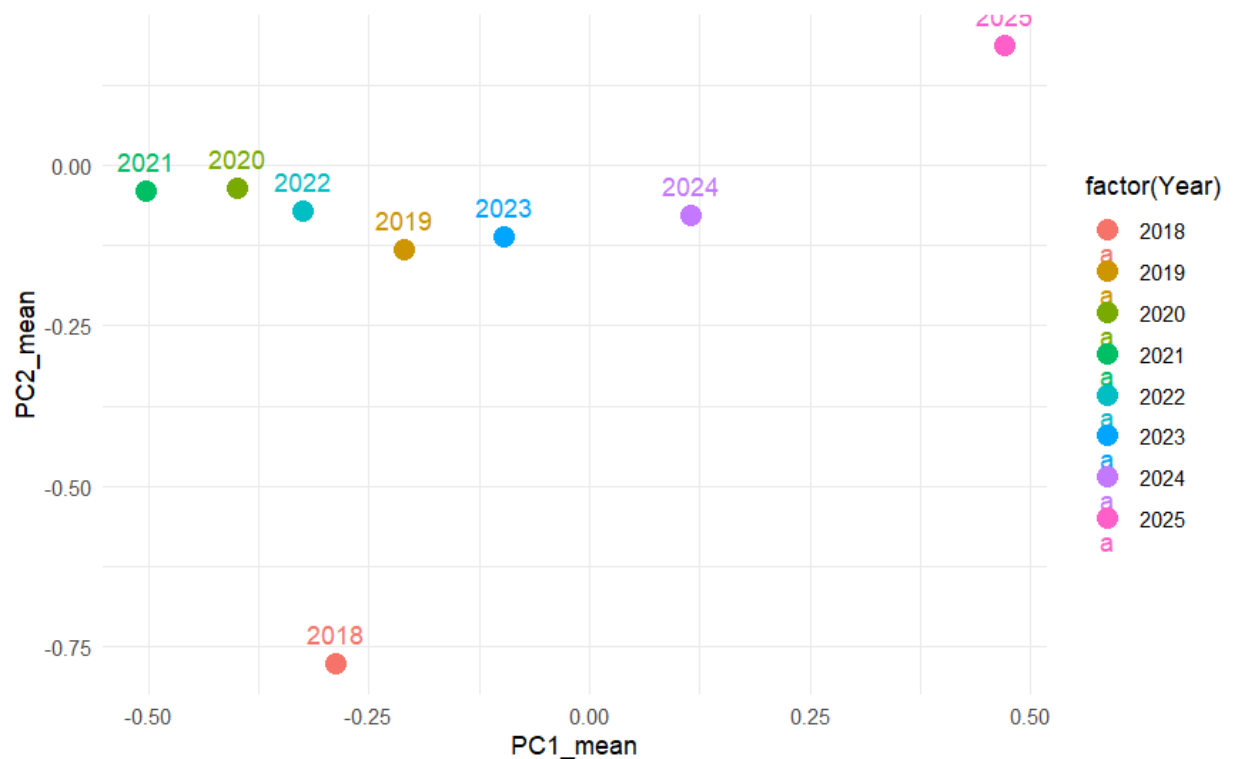
Smut genres saw almost no increase from what would have been expected from previous years, and some even saw decreases. Because we cannot identify individual fictions as being part of the spike or expected fictions, and genres are not exclusive, we cannot directly examine the proportions of the spike fictions, but we can examine the absolute increase in the number of fictions of each genre, see fig 2. Fig 2 uses a linear model trained on the data prior to March 2025 and measures the accuracy of that model and how much deviation there is after March. Variable is the genre, $r_squared$ and $adj_r_squared$ measure how well the linear model predicts that genre, $mean_percent_inc$ measures the proportional increase in that genre from what the model predicted (eg 1 means no difference, 2 means it doubled from what was expected), $mean_absolute$ increase measures the mean difference between the number of fictions published and the number of fictions that the model predicted would be published. Most of the genres have an $adj_r_squared$ of around 0.8, which indicates that a linear model is a good fit. Those genres with a substantially lower $adj_r_squared$, such as Sports with a value of 0.319, are much less popular than other genres which leads to increased variation and inconsistent month over month growth. One reason that erotic content saw no change while other genres saw so much could be content filters. If this increase in fictions is truly caused by increased LLM adoption, the fact that many LLMs restrict sexually explicit content would result in this exact finding.

Fig 2

variable	r_squared	adj_r_squared	mean_percent_inc	mean_absolute_inc
Fantasy	0.9188387	0.9176625	1.3846039	164.76846076
Action	0.9161801	0.9149653	1.3605680	155.59069416
Adventure	0.9074837	0.9061428	1.3587654	132.20970825
Drama	0.8680154	0.8661026	1.8424514	130.49391348
Mystery	0.9025663	0.9011542	1.9475619	117.20623742
Psychological	0.8840818	0.8824018	2.0552991	111.13521127
Romance	0.8749586	0.8731464	1.4457966	105.40005030
Supernatural	0.9041409	0.9027517	1.4174991	83.92500000
Tragedy	0.8859656	0.8843129	1.8775298	82.86524145
Comedy	0.7655804	0.7621831	1.4059658	59.80970825
Mature	0.8507425	0.8485794	1.4061752	58.93546278
Sci-fi	0.8750145	0.8732031	1.5488122	57.66292757
Slice of Life	0.8985218	0.8970511	1.3489320	54.05482897
Horror	0.8668327	0.8649027	1.7800069	48.29386318
Adult	0.8939412	0.8924042	1.2367181	38.63214286
Isekai	0.8715033	0.8696410	1.1853751	34.06956740
School Life	0.7842973	0.7811712	1.3300273	28.76649899
Historical	0.6471918	0.6420786	1.7889850	25.87867203
Seinen	0.7498683	0.7462432	1.6165908	22.16705231
Martial Arts	0.7970503	0.7941090	1.2417673	21.41267606
LitRPG	0.8034274	0.8005786	1.1981252	20.40563380
Boys Love	0.2005020	0.1889150	1.6167632	14.98853119
Smut	0.8279769	0.8254838	1.0864940	8.21257545
Sports	0.3301180	0.3187640	1.9356475	5.27085272
Josei	0.1942321	0.1825543	1.3366673	2.38772636
Mecha	0.6369138	0.6315743	1.1196125	2.03670240
Gender Bender	0.5120448	0.5049730	1.0174095	0.92525151
Harem	0.7518687	0.7482726	1.0010134	-0.01574447
Ecchi	0.7154040	0.7112794	0.9665679	-1.93586519
Fanfiction	0.7504372	0.7468203	0.9766016	-3.95573441
Girls Love	0.8238290	0.8212758	0.9527531	-4.15045272

Additionally, the style of the chapters, as measured by the relative frequencies of words, shifted substantially compared to previous years as shown by fig 3. This graph shows the centroid of all of the fictions per year, with the x and the y being linear combinations of the relative word frequencies that together represent 31% of the variation in them. 31% is rather low, and although the data for 2025 is certainly different from what came before, it is not enough on its own to say that the style of writers has significantly changed in this year.

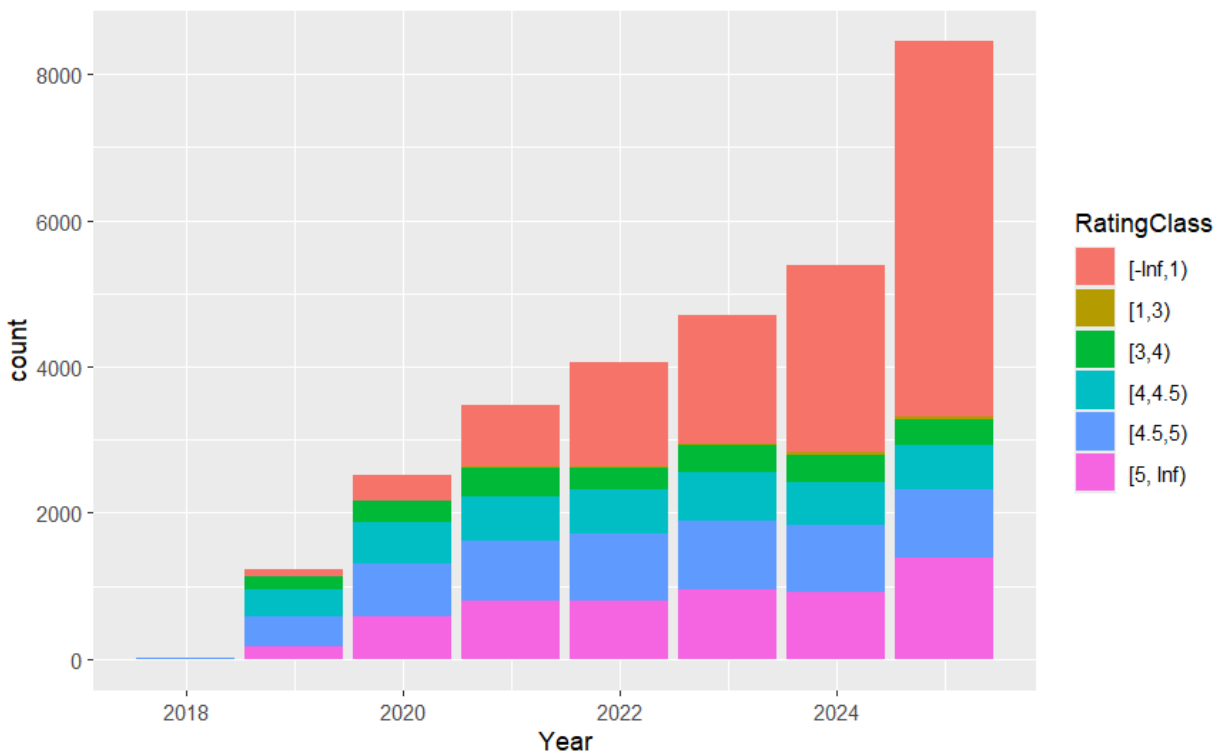
Fig 3



A large majority of the fictions published on the site have 0 star ratings, indicating that no one has rated them. Fictions can only be rated 1-5, and a smaller but still large portion of the fictions have 5 star ratings. Both 0 and 5 star ratings grow over time, while other rating types do not seem to increase. Only 0 star ratings grew proportionally, while 5 star rating proportions stayed constant. This is consistent with what would be expected from an increase in authors due to LLMs decreasing the difficulty of writing, which would not coincide with an increase in the amount of readers able and willing to rate those fictions. Another explanation would be that fictions produced using LLM

assistance are of substantially lower quality compared to traditionally produced fictions. It should be noted that the older a fiction is, the more time it has had to potentially gather ratings, which would somewhat bias these results. These findings give evidence against the hypothesis that this sudden increase in fictions was simply organic growth, as you would expect the proportion of unrated fictions to remain constant, or at least not grow as much if readers were growing as quickly as writers are.

Fig 4



Conclusion

This paper analyzes a variety of trends on Scribblehub and comes to the conclusion that they support a substantial increase in the amount of LLM assisted or generated fictions beginning in March 2025 more than any alternative explanation. The ratings patterns are against the hypothesis that the growth is natural, and the length of the

effect is inconsistent with the idea that this is driven by some type of event. Although the evidence supports the LLM hypothesis, this study is observational and cannot say with certainty that the changes in the number of fictions published was because of LLMs. The genre difference also supports the LLM hypothesis, as other than content restrictions, this is difficult to explain.

This study was only conducted on one online writing website, and its findings cannot necessarily be extrapolated to all forms of online writing. Future research might extend this analysis to other writing websites, such as Royalroad or Webnovel. This increase in LLM adoption, even in low incentive environments such as this one, will decrease discoverability for new authors, and pollute future training data for LLMs. Only time will tell how the adoption of LLMs will affect the quality and variability of fiction.

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